

Single Molecule Surface Enhanced Raman Scattering in a Single Gold Nanoparticle-Driven Thermoplasmonic Tweezer

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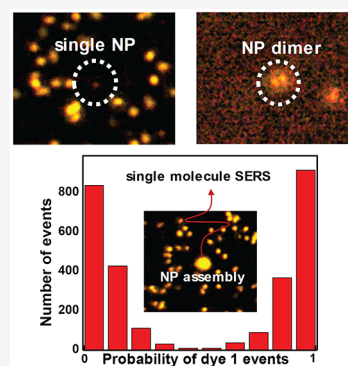


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Supporting Information

ABSTRACT: Surface enhanced Raman scattering (SERS) is optically sensitive and chemically specific to detect single-molecule spectroscopic signatures. Facilitating this capability in optically trapped nanoparticles at low laser power remains a significant challenge. In this letter, we show single molecule SERS signatures in reversible assemblies of trapped plasmonic nanoparticles using a single laser excitation (633 nm). Importantly, this trap is facilitated by the thermoplasmonic field of a single gold nanoparticle dropcasted on a glass surface. We employ the bialynte SERS technique to ascertain the single molecule statistical signatures and identify the critical parameters of the thermoplasmonic tweezer that provide this sensitivity. Furthermore, we show the utility of this low power ($\approx 0.1 \text{ mW}/\mu\text{m}^2$) tweezer platform to trap a single gold nanoparticle and transport assembly of nanoparticles. Given that our configuration is based on a dropcasted gold nanoparticle, we envisage its utility to create reconfigurable plasmonic metafluids in physiological and catalytic environments and to be potentially adapted as an *in vivo* plasmonic tweezer.



Optical confinement and interrogation of single molecules in a fluid remains one of the vital goals of nanoscience and technology.^{1–6} This letter discusses how single molecule surface enhanced Raman scattering (SERS) can be performed in a single gold nanoparticle-driven thermoplasmonic trap of plasmonic particles. Fluorescence and Raman scattering have been extensively used to study molecules in a fluid. Generally, fluorescence is sensitive enough to detect single molecules in a fluid, but Raman scattering remains a weak process due to low optical cross sections.² An important way to overcome this problem is to use SERS,^{7–11} a spectroscopic method that is sensitive enough to detect single molecules in a fluid phase.^{7,12–18} An important prerequisite for single molecule SERS (SM-SERS) detection is the enhanced electromagnetic fields.^{19–22} Plasmonic structures have been extensively designed and utilized for near-field interaction with the molecules,^{23–27} and thereby influencing their optical emission not only in terms of their intensity^{28–30} but also in terms of their directivity.^{31,32}

To perform single molecule SERS, a large electromagnetic field enhancement is necessary.^{33–35} An efficient way to boost this enhancement is to create assemblies of plasmonic nanoparticles.³⁶ In a fluid phase, a majority of plasmonic colloidal assemblies tend to form aggregates that cannot be reversed. This is a major disadvantage in micro- and nanofluidic environments, where irreversible aggregation of colloids can lead to clogging of fluidic channels.^{37–39}

In the context of optofluidic molecular detection, optical^{40–43} and plasmonic^{44–51} trapping can be used to create reversible assemblies of SERS active colloids,⁵² which can avoid aggregation and clogging. Furthermore, such

colloidal assemblies can be transported to a desired location within the fluidic channel.^{53–58} Also, optothermal effects can be harnessed to trap and transport various micro- and nano-objects^{59–62} including single macromolecules.⁶³

Until now, most of the optical trapping-based SERS studies have used at least two lasers (one high power laser for trapping, and a lower power for Raman excitation) in their experiments^{64–66} or have created large-scale plasmonic colloids^{13,57} with a single laser on a metal film. Such configurations hinder low-power laser operation, simplification, and miniaturization of single molecular SERS tweezers for micro- and nanofluidic applications.

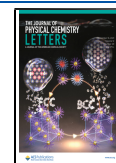
There is an imperative to create reversible assemblies of SERS colloids on a glass surface, down to the limit of a dimer or a few nanoparticles with a single, low-power laser beam that can be harnessed for single molecule SERS in a microfluidic environment. This motivated us to look for alternate trapping and tweezing strategies with the following requirements:

- Low power, single-excitation optical control of reversible nanoassembly of plasmonic colloids with single-molecule SERS sensitivity.
- Reversible assembly down to the single nanoparticle limit.

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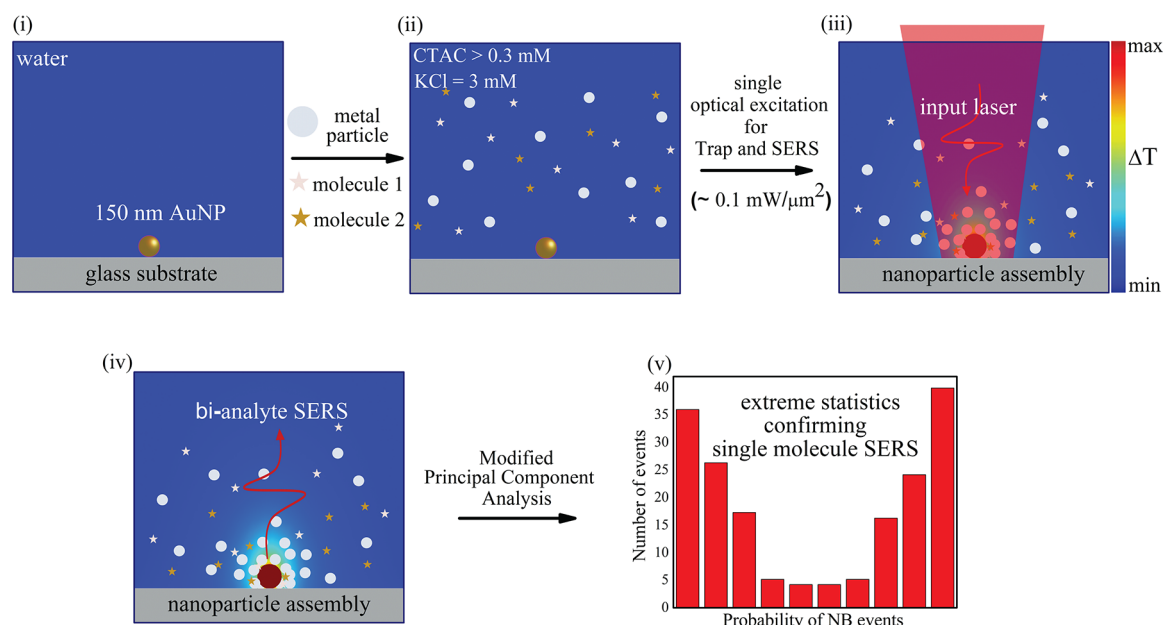


Figure 1. Schematic of the experimental configuration to perform SM-SERS in a low power, single nanoparticle driven thermoplasmonic tweezer. (i) A single 150 nm AuNP dropcasted on a glass substrate to be operated as an anchor particle for trapping. (ii) The AuNP was surrounded by a solution containing CTAC and KCl mixed with molecules and plasmonic particles (AuNPs or Lee-Meisel AgNPs). (iii) A low power density laser source ($\approx 0.1 \text{ mW}/\mu\text{m}^2$) was used to heat the AuNP to achieve reversible assembly of plasmonic particles. (iv) SERS was performed on the molecules in the assembled plasmonic particles using the same laser source. (v) Modified principal component analysis was performed to statistically confirm the SM-SERS signature.

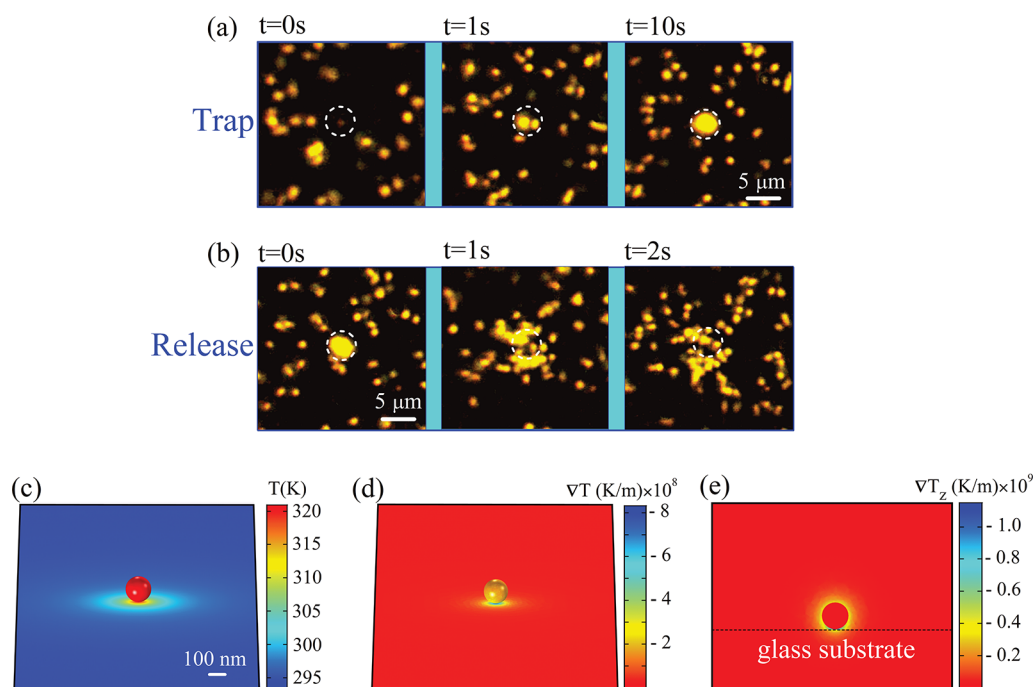


Figure 2. Reversible trapping of plasmonic particles in the low power thermoplasmonic tweezer. Time series dark field images of (a) assembly of 250 nm AuNPs and (b) disassembly of the same AuNPs when the laser was turned off. The dotted, white circle represents the location of the anchor 150 nm AuNP. Excitation at 633 nm laser with a power density of $0.08 \text{ mW}/\mu\text{m}^2$. (c) Calculated temperature distribution around a 150 nm AuNP placed on a glass substrate and surrounded by water. Excitation at 633 nm laser with a power density of $0.08 \text{ mW}/\mu\text{m}^2$. (d) Calculated temperature gradient ∇T for the same geometry. (e) Calculated temperature gradient ∇T in the z -plane. See Movie 1 for trapping and release of 250 nm AuNPs.

- Reversible assembly that can be achieved within a few seconds.

In this paper, we show how all the above requirements can be met by a new strategy of single-nanoparticle-driven

thermoplasmonic tweezer. Specifically, we show how SM-SERS can be performed in a thermoplasmonic trap driven by a single, dropcasted gold nanoparticle. This dropcasted nanostructure acts as an anchor trap to create an assembly of

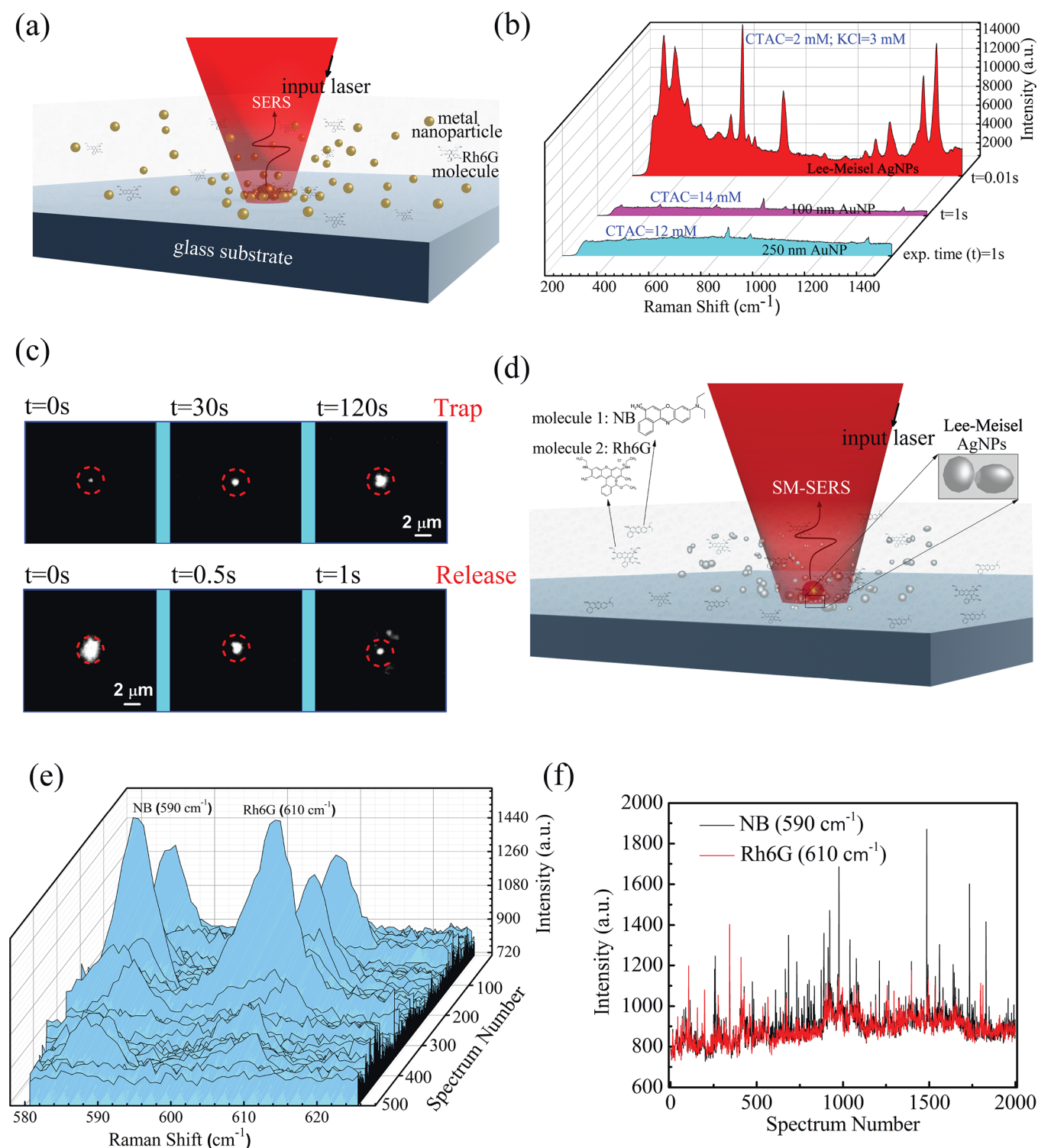


Figure 3. Optimization of experimental configuration for SM-SERS. (a) Schematic of the SERS performed for different plasmonic particles (250 and 100 nm AuNPs, and Lee-Meisel AgNPs) with Rh6G molecules. (b) Comparison of the SERS spectra at the same input power using 250 nm AuNPs, 100 nm AuNPs, and Lee-Meisel AgNPs (with varied concentrations of CTAC and KCl). (c) Time series images of assembly and disassembly of the Lee-Meisel AgNPs. (d) Schematic of SM-SERS using bianalyte method at 633 nm laser. The trapping configuration included the following: 150 nm anchor AuNP, Lee-Meisel AgNPs in solution, NB and Rh6G molecules in the presence of CTAC (2 mM) and KCl (3 mM). (e) Waterfall plot of 500 bianalyte SERS spectra recorded from a solution containing 5 nM of both NB and Rh6G molecules. The integration time for a single spectrum was 0.3 s, and the dwell time between two consecutive spectra was 1 s. (f) SERS intensity variation of 590 and 610 cm⁻¹ peaks of NB and Rh6G molecules, respectively, as a function of spectrum number. See Movie 5 and Movie 6 for trapping and release of Lee-Meisel AgNPs, respectively.

plasmonic particles which can be further utilized for SM-SERS studies. By utilizing the bianalyte SERS technique,⁶⁷ we

provide experimental proof of the SM-SERS signature in such thermoplasmonic traps. Furthermore, we show how such

nanoparticle-based thermoplasmonic traps can be utilized for single nanoparticle trapping and optical transport of nanoparticle assemblies, all at a very low laser power density.

A schematic of the experimental configuration is shown in Figure 1. A single 150 nm diameter gold nanoparticle (AuNP) was dropcasted on a glass substrate to work as an anchor particle to create an assembly of plasmonic structures. The AuNP was surrounded with an aqueous solution of plasmonic particles, molecules, surfactant cetyltrimethylammonium chloride (CTAC) and potassium chloride (KCl). The plasmonic particle assembly provides the electromagnetic enhancement for the SM-SERS.⁷ The CTAC and the KCl were used as assisting agents for thermoplasmonic trapping and SM-SERS, respectively. We discuss their role later in the manuscript. A single, low-power-density laser was used to simultaneously heat the anchor metal nanoparticle⁶⁸ for thermoplasmonic trapping of plasmonic particles, and as an excitation source for SERS. For SM-SERS validation, we used the bianalyte technique. Two types of molecules were added in the solution after which modified principal component analysis (MPCA) was performed to statistically confirm the SM-SERS signature.⁶⁹ See Supporting Information (SI), section S1 for a detailed schematic of the optical microscope, and SI section S2 for discussion on the mechanism of the reversible trap.

First, we discuss the experiments on reversible trapping of 250 nm AuNPs in a CTAC solution. Figure 2a,b show the trapping and release sequence of 250 nm AuNPs. A 150 nm AuNP was excited by focusing a 0.74 mW laser of wavelength 633 nm at a 3.34 μm spot size by using a 50 $\times$, 0.50 NA objective lens. The AuNP was surrounded with an aqueous solution having 12 mM CTAC. As the laser was turned on, the plasmonic particles were seen to be trapped near the anchor AuNP and within 10 s, an assembly of size greater than 5 μm was achieved. The assembly disappears within a few seconds after turning off the laser. This process can be reproduced over multiple cycles. See SI Movie 1 for trapping and release of 250 nm AuNPs.

In the absence of the CTAC, the trapping of plasmonic particles was not observed at the same input power density.⁵⁷ The particles get scattered away because of the large scattering force when they enter the excitation volume (see Movie 2). Furthermore, in the absence of anchor AuNP and CTAC, the trapping of plasmonic particles was not observed upon the excitation of only glass substrate. This shows that conventional optical trapping at such a low power density cannot trap the plasmonic particles because of the absence of a sufficient temperature gradient in the system (see Movie 3). Assembly of nanoparticles of size as large as 400 nm were also created efficiently at low power density, using the single nanoparticle driven thermoplasmonic trapping (see Movie 4). A slightly more efficient trapping at 532 nm was achieved with a laser source of 532 nm which is nearer to the absorption maxima of the anchor particle. See SI sections S3 and S4 for the details on the measurement of scattering cross-section and calculation of absorption cross-section of 150 nm AuNP placed on a glass substrate and surrounded with water, and trapping experiments performed with a 532 nm laser. Note that our thermoplasmonic tweezer is mainly driven by a single anchor AuNP on a glass substrate, in contrast to metal film-based tweezers.⁷⁰

To further understand the optothermal effects⁷⁰ in our tweezer, we performed finite element method simulations (COMSOL). The refractive index of gold was taken from ref 71 (see SI section S5 for details on numerical calculations).

Figure 2c shows the calculated temperature distribution of a 150 nm AuNP placed on a glass substrate and surrounded with water. A temperature increment of ≈ 27 K was obtained by exciting the AuNP with a laser of power density 0.08 mW/ μm^2 . In addition, there was a temperature gradient, both in-plane and out-of-plane of our geometry, because of the heating of AuNP as shown in Figure 2d,e, respectively. This gradient has an important connection to CTAC in the solution. The plasmonic particles in the fluid and the anchor particles were coated with a positive CTAC double layer because of the addition of CTAC in the solution.⁷² Since, the concentration of CTAC is greater than the critical micelle concentration (0.3 mM), the solution also contains CTAC micelles and negatively charged chlorine ions. The temperature gradient creates a thermophoretic movement⁷³ of CTAC double layer coated plasmonic particles, and CTAC micelles and Cl^- ions present in the solution, and facilitates the reversible trapping of a metallic structure. The particles get trapped near the heated anchor AuNP because of the balancing of the thermogradient force between negatively charged Cl^- ions and positively charged CTAC micelles, and the electrostatic repulsive force between the positive coating of the CTAC double layer on the anchor particle and the CTAC double-layer coated metal particles.

Having shown the formation of large scale assembly, next we show how the assembly can be utilized to perform SERS of rhodamine 6G (Rh6G) molecules and compare its performance for various plasmonic particles. A schematic of the experimental design is shown in Figure 3a. A reversible assembly of plasmonic particles (250 or 100 nm AuNPs or Lee-Meisel AgNPs) was achieved by heating a 150 nm AuNP. The anchor AuNP was illuminated with a water immersion 1.2 NA, 60 $\times$ objective lens by focusing a 0.80 mW, 633 nm laser to a spot size of 2.70 μm (For this optical excitation, the temperature distribution, and gradients are shown in SI section S6). After an assembly formation of about the spot size, SERS spectroscopy of Rh6G molecules was performed in the backscattering configuration using the same excitation laser used for the trapping purpose. Figure 3b compares the SERS spectrum collected from 10^{-6} M Rh6G molecules in the assembly of 250 nm AuNPs, 100 nm AuNPs, and Lee-Meisel AgNPs. As can be seen from the SERS spectra Figure 3b, the SERS sensitivity was superior for Lee-Meisel AgNPs in the presence of CTAC and KCl. From the past work, it is well-known that Lee-Meisel AgNPs are excellent substrates for achieving large SERS enhancement,^{7,74} when a small amount of KCl is added to the solution.^{75–77} Therefore, Lee-Meisel AgNPs were first mixed with KCl (3 mM) and then with Rh6G molecules by following a half–half dilution method.⁷⁸ Again using a half–half dilution method, a low concentration of CTAC (2 mM) was then added to the solution containing molecules and AgNPs.

Figure 3c shows the trap and release of a large scale assembly of Lee-Meisel AgNPs using a nanoparticle driven thermoplasmonic trap. An assembly of a few micrometers was formed within tens of seconds after the anchor particle was illuminated with the laser source. The assembly disappears very fast after the laser is turned off. See Movie 5 and Movie 6 for trapping and releasing of Lee-Meisel AgNPs, respectively, and SI section S7 for details on the preparation of Lee-Meisel AgNPs.

To push the limit of single nanoparticle driven thermoplasmonic trap, we performed SM-SERS spectroscopy on the

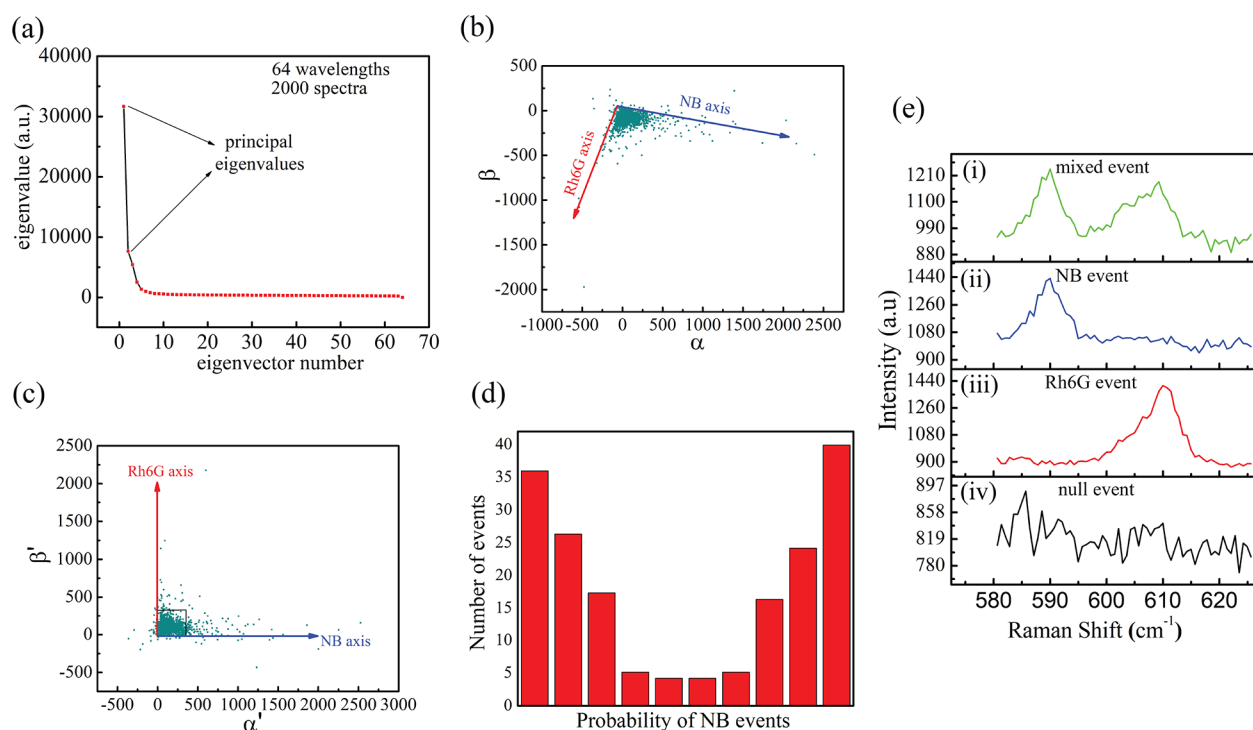


Figure 4. Modified principal component analysis of bianalyte SM-SERS. (a) Eigenvalues obtained from PCA of 2000 spectra consisting of 64 wavelengths containing the 590 and 610 cm⁻¹ peaks of NB and Rh6G, respectively. (b) Distribution of the covariant matrix coefficients (α and β) obtained after PCA showing two Rh6G and NB axes. (c) Distribution of the transformed matrix coefficients α' and β' showing two orthogonal Rh6G and NB axes. (d) Probability of NB events plotted using data from panel c after removing the mixed dye events shown in the black region in panel c. (e) Four possible events: (i) Mixed event obtained near the center of the histogram (d); (ii) NB event obtained near the region of probability of NB event = 1; (iii) Rh6G event obtained near the region of probability of NB event = 0; (iv) no event obtained when probabilities of both NB event and Rh6G event are 0.

assembly of trapped Lee-Meisel AgNPs using bianalyte technique, the details of which can be found elsewhere.^{13,67,79}

Figure 3d shows the schematic of the experimental configuration to perform bianalyte SERS experiments using two types of molecules, which compete for the same hotspot.⁶⁷ A 3 mM KCl solution was added with Lee-Meisel AgNPs and was left for 10 min. The molecules were then added in the solution using the half–half dilution method.⁷⁸ Finally, CTAC was added to the solution again using the half–half dilution method. The solution contains two types of molecules, Nile Blue (NB) and Rh6G, with a final concentration of 5 nM. As discussed earlier, the illuminated anchor AgNPs drive the assembly, and SERS was performed from the assembly in the presence of CTAC and KCl. See SI section S8 for bianalyte SERS experiments performed adding KCl in the solution assembly.

Next, we discuss the obtained SERS spectra. A waterfall plot of 500 spectra containing closely spaced NB SERS peak 590 cm⁻¹ and Rh6G SERS peak (610 cm⁻¹) is shown in Figure 3e. The exposure time for each spectrum was 0.3 s, and dwell time between two consecutive spectra was 1 s. The variation in the peak intensity can be seen clearly. Figure 3f shows the occurrence and variation in the intensity of both the 590 cm⁻¹ peak of NB and the 610 cm⁻¹ peak of Rh6G molecules as a function of spectrum number. These fluctuating individual events suggest that the single nanoparticle driven thermoplasmonic trap can be used to perform SM-SERS.

To statistically confirm the single molecule SERS signature, we performed modified principal component analysis (MPCA)^{13,69,79} on 2000 spectra collected (Figure 3e) from

the solution containing both NB and Rh6G molecules. Figure 4a shows the plot of calculated eigenvalues and eigenvectors, in which the first two elements are the principal eigenvalues. Figure 4b shows the scatterplot of two covariant matrix coefficients α and β , with two axes shown as NB and Rh6G. Each data point in the scatterer plot represents an event. The reoriented scatterer plot (α' and β') in Figure 4c shows two orthogonal axes, the Rh6G and NB axes, where events on both axes represent pure events of SERS from only one type of molecule. Events in between the two axes represent mixed events which are signals from both types of molecules. To calculate the frequency of single molecule events, a histogram was plotted in Figure 4d where the frequency of only the NB molecule was plotted after removing noisy mixed events as shown by a black box in Figure 4c. The extreme left part of the histogram represents the events where SERS from only Rh6G molecules was recorded. The spectrum in Figure 4e(iii) shows such a type of an intense pure event from only Rh6G molecules. The center part of the histogram shows the mixed events from both types of molecules and one such event is shown in Figure 4e(i). The extreme right part of the histogram displays the probability of only NB molecule events and one such event is shown in Figure 4e(ii). Along with these three events, a relatively less probable null event was also observed, which is shown in Figure 4e(iv). For further details on the MPCA analysis, see SI section S9.

Can our tweezer platform assemble and transport plasmonic structures? Can such a platform trap single nanoparticles at the low laser power densities? To address these questions, further experiments were performed as shown in Figure 5. First, we

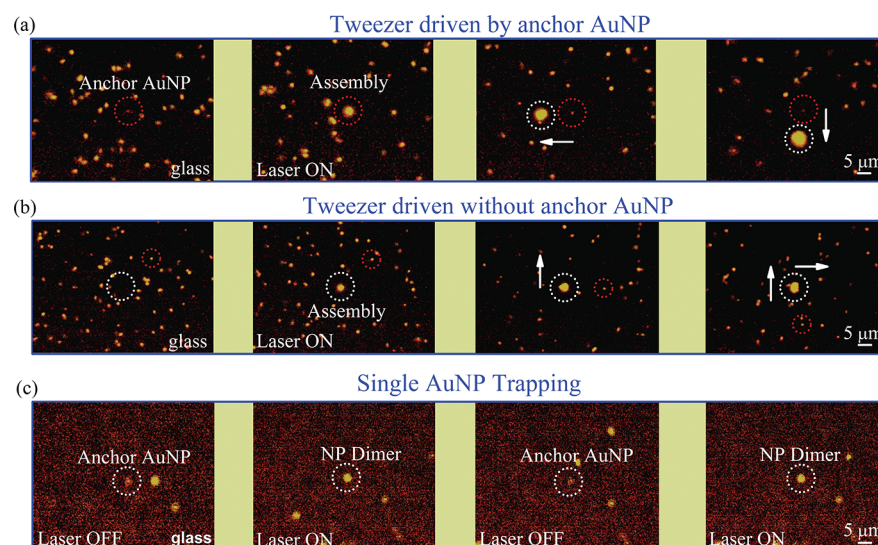


Figure 5. Versatility of nanoparticle driven thermoplasmonic tweezer. (a) Tweezer driven by anchor AuNP. Time series dark field images of assembly and its translation along two axes. (b) Tweezer driven without anchor AuNP. Time series dark field images of assembly and its translation along two axes. Note that this assembly process is stochastic and takes a longer duration compared to the previous case. (c) Single particle trapping and release. A reversible dimer formation was observed. Excitation wavelength: 633 nm at a power density of 0.08 mW/μm², anchor AuNP: 150 nm, trapped plasmonic particles: 250 nm AuNPs, for all the observations. See [Movies 7, 8, and 9](#) related to the above data.

trapped and transported an assembly of 250 nm AuNPs along two different axes, as shown in [Figure 5a](#). We performed this transport by holding the trap at its position and moving the motorized-stage of the optical microscope. (See [Movie 7](#) showing the working of a tweezer driven by the anchor AuNP.) An interesting aspect of this transport is that once assembled, the plasmonic particles do not need the assistance of the anchor AuNP for further trapping. This may be due to the temperature increment and gradient facilitated by the assembly and hence acting as a “self-sufficient” trap. For a fixed input power density, the temperature increment in the case of dimer and trimer of particles is more than a single particle (see [SI section S10](#)). This shows that the assembly of nanoparticle can provide a sufficient temperature gradient in the system to trap plasmonic particles even in the absence of an anchor AuNP. Though the temperature increases after formation of the assembly, no photodamage was observed.

Second, we show that the assembly can also be created without the assistance of an anchor NP ([Figure 5b](#)). Although we could trap and transport the assembly of plasmonic particles, we found the trapping process to be stochastic and can take much longer times when compared to the anchor AuNP based tweezers ([Figure 5a](#)). [Figure 5b](#) shows the formation of the assembly when the laser was illuminated on a glass substrate. Initially, a single particle comes under the excitation volume and gets pushed toward the glass substrate because of scattering force, and it stays there for a finite duration. Within that time, the temperature of the particle increases, which leads to the formation of a thermal gradient, and hence a trap. See [Movie 8](#) showing the working of tweezers driven without anchor nanoparticles.

Finally, we show how a single 250 nm plasmonic particle can also be trapped using a single, anchor AuNP, thus forming a dynamic, plasmonic dimer. [Figure 5c](#) shows the reversible trapping and release of a single 250 nm AuNP. Turning off the laser results in the release of the trapped particle from the anchor location. Turning on the laser again, further traps the

nanoparticle thus confirming the reversibility. See [Movie 9](#) showing single nanoparticle trapping.

To conclude, we have shown how a single dropcasted gold nanoparticle on a glass substrate can be utilized as a thermoplasmonic tweezer to reversibly assemble plasmonic nanoparticles for SM-SERS experiments. By employing the bialyte SERS technique we have revealed the single-molecule spectral dynamics and its extreme statistical signatures in the assembly of nanoparticles. We have emphasized the relevance of surfactant and salt solution in creating a conducive environment for trapping and SM-SERS at low laser powers ($\approx 0.1 \text{ mW}/\mu\text{m}^2$). Furthermore, we have revealed the capability of the tweezer to not only trap single plasmonic nanoparticles, but also to transport plasmonic nanoparticle assembly to a desired location. Such an assembled plasmonic entity can be potentially harnessed as components of plasmonic meta-fluid. Given that our tweezer platform is based on a colloidal gold nanoparticle, it can be chemically functionalized for targeted trapping of nano-objects including biomolecules. We envisage its utility as a flexible trapping and transport platform *in vivo*, and anticipate further exploration on colloidal plasmonic tweezers sensitive to single-molecule reactions in physiological conditions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpclett.1c03450>.

Experimental setup, mechanism of trap, scattering and absorption cross sections, details on FEM simulations, details on MPCA, controlled experiments on bialyte SERS with 633 nm laser ([PDF](#))

Reversible trapping of 250 nm AuNPs ([MP4](#))

Effect of absence of CTAC on trapping ([MP4](#))

Effect of absence of anchor AuNP and CTAC on trapping ([MP4](#))

Reversible trapping of 400 nm AuNPs ([MP4](#))

Trapping of Lee-Meisel AgNPs ([MP4](#))

Release of Lee-Meisel AgNPs assembly (MP4)
Working of tweezer driven by anchor AuNP (MP4)
Working of tweezer driven without anchor AuNP (MP4)
Trapping of a single 250 nm AuNP (MP4)
Reversible trapping of 250 nm AuNPs using 532 nm laser (MP4)
Unprocessed movie of reversible trapping of Lee-Meisel AgNPs (MP4)

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Notes

The authors declare no competing financial interest.

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